修 士 論 文 の 和 文 要 旨

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論 文 題 目	Bayesian fused lasso regression modeling for binary data via horseshoe prior (二値データにおける馬蹄事前分布によるベイジアン連結 lasso 回帰モデリング)		

要　　旨

近年，多様化を極めるデータから得られる情報や知見の需要は高まり続けており，データに内在する疎構造を抽出する手法である L_1 型正則化法は注目を集めている．代表的な手法としては回帰係数と連続する回帰係数の差に対してそれぞれの絶対値 (L_1 ノルム) の和を正則化の罰則項として課す連結 lasso がある．L_1 ノルムによる罰則項により，連結 lasso は回帰係数と連続する回帰係数の差の一部を 0 と推定し，変数選択と回帰係数の併合の機能を実現する．加えて，量的データの解析法としての L_1 型正則化法を，0 か 1 かをとる二値データの分析法であるロジスティック回帰モデルへ拡張した手法も広く活用されている．例えば連結 lasso を用いたロジスティック回帰モデルはスペクトルデータや医用画像データの分析に用いられている．

ベイズ統計学の視点から捉えると，L_1 型正則化法は回帰係数にラプラス事前分布を仮定することに対応する．連結 lasso をベイズ的に解釈した手法であるベイジアン連結 lasso は，ラプラス分布を回帰係数と連続する回帰係数の差に仮定することで両者を 0 に向かって縮小する．しかし，ラプラス分布は 0 でないはずの回帰係数の差まで大きく 0 に縮小する傾向にある．縮小せずに残しておくべき回帰係数の差が 0 に近づくことによって，モデルがデータに内在する構造を捉えきれなくなる可能性が生じる．

本論文では，馬蹄事前分布によるベイジアン連結 lasso を用いたロジスティック回帰モデルを提案する．連続する回帰係数の差に馬蹄事前分布を仮定することで過度な縮小を防ぎ，非 0 と推定されるべき部分を残したモデリングを実現することを目指す．さらに，事前分布および尤度の階層表現によりギブスサンプラーを構成し，簡便なパラメータ推定アルゴリズムを提案する．また，提案手法の有効性を人工データによるシミュレーションにより検証する．検証の結果，多くの場合において提案手法は従来手法に比べて正確に連続する回帰係数の差を捉え，より高い予測性能を持つことが確認された．

令和四年度　修士論文

Bayesian fused lasso regression modeling for binary data via horseshoe prior

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1 Introduction

L_1-norm regularization is one of the methods which enable the analysis of data with sparse structure. A representative L_1-norm regularization method is Lasso (Tibshirani, 1996). Lasso performs variable selection by imposing the sum of the absolute values (L_1-norms) of regression coefficients as a penalty term on log-likelihood function. Tibshirani et al. (2005) extended lasso to the method for analyzing data with group structure, called fused lasso. Fused lasso estimates some of the regression coefficients and the successive differences of them as zero with a regularization. These features of the estimation enable variable selection and fusion of variables. Fused lasso can be used for data whose variables are some sense ordered and successive variables have the same role for prediction.

These methods mainly consider linear regression models which can analyze quantitative data. For qualitative data such as binary data, logistic regression model is often used. Tibshirani (1996) also suggested the expansion of lasso to the logistic regression model. It can be used as the statistical method in genome analysis and the prediction of corporate bankruptcy (Wu et al. (2009); Pereira et al. (2016)). There are also several studies of logistic regression with fused lasso. For example, Yu et al. (2015) used the logistic regression model with fused lasso for spectral data. Xin et al. (2014) also proposed the fused lasso for binary data to a diagnosis of Alzheimer's disease.

As the frequentist methods of regularization are developing, many Bayesian approaches also have been proposed. Park and Casella (2008) proposed Bayesian lasso which shrinks regression coefficients by assuming Laplace prior on them. Kyung et al. (2010) proposed Bayesian fused lasso, which expanded Bayesian lasso by assuming Laplace prior on regression coefficients and their successive differences. Both of these methods adopt a Gibbs sampling for the parameter estimation using the hierarchical expression of a Laplace prior by a scale mixture of normals (Andrews and Mallows, 1974). Genkin et al. (2007) proposed logistic regression with Bayesian lasso and applied it to the document classification problem. Betancourt et al. (2017) proposed logistic regression model with fused lasso for analyzing data from a finan-

cial trading network. However, a Laplace prior tends to shrink its targets such as the regression coefficients and their successive differences too much. Therefore, a Laplace prior often leads to fail to capture the structure of data. To overcome this problem, horseshoe prior (Carvalho et al., 2010) is widely used. Horseshoe prior has an infinite spike and strongly shrink elements to zero and a heavy tail. This feature enables strong shrinkage of the small elements and prevent the over-shrinkage of large and important elements. Makalic and Schmidt (2015) proposed logistic regression model assuming horseshoe prior on regression coefficients. However, existing studies for logistic regression model assumes horseshoe prior only on regression coefficients.

In this paper, we propose a Bayesian logistic regression model assuming Laplace prior on regression coefficients and horseshoe prior on their successive differences. To estimate the parameters of the model, we develop the Gibbs sampler by using data-augmentation method with Pólya-Gamma distribution (Polson et al., 2013).

The remainder of this paper is organized as follows. Section 2 introduces the L_1-norm regularization methods. Section 3 describes the Bayesian sparse modeling methods. In Section 4, we propose a logistic regression model using Bayesian fused lasso with horseshoe prior and the sampling algorithm for the parameter of the model. Section 5 shows Monte Carlo simulations and the results. In Section 6, we conclude this paper and discuss the future work.

2 L_1 regularized logistic regression

In this section, we introduce L_1-norm regularization methods which utilize the absolute values of regression coefficients as a penalty term. Regularization is known as a way to prevent overfitting, and then L_1-norm regularization also enables to capture the sparse structure in data. Especially, variable selection and variable fusion are well-known benefits of L_1 regularization. Variable selection is an effective way of choosing the important variables from large number of other ones. Variable fusion is often used to capture the group structure of variables.

We start this section with the introduction of regularized likelihood method and describe logistic regression models with lasso and fused lasso.

2.1 Regularized likelihood method

Suppose that we have the binary response variables $y_1, y_2, \cdots, y_n \in \{0, 1\}$ and p-dimensional vector of explanatory variables $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_n$. Logistic regression model is formulated as

$$\Pr(y_i = 1 \mid \boldsymbol{x}_i) = \frac{1}{1 + e^{-(\beta_0 + \boldsymbol{x}_i^T \boldsymbol{\beta})}}, \quad i = 1, \ldots, n, \tag{2.1}$$

where $\Pr(y_i = 1 \mid \boldsymbol{x}_i)$ is the probability that y_i is 1 given $\boldsymbol{x}_i$. β_0 is an intercept and $\boldsymbol{\beta} = (\beta_1, \beta_2, \cdots, \beta_p)^T$ is a p-dimensional regression coefficient vector.

Then, the likelihood function is expressed as follows:

$$f(\boldsymbol{y} \mid \boldsymbol{X}; \beta_0, \boldsymbol{\beta}) = \prod_{i=1}^{n} f(y_i \mid \boldsymbol{x}_i; \beta_0, \boldsymbol{\beta}), \tag{2.2}$$

where $\boldsymbol{y} = (y_1, y_2, \ldots, y_n)^T$ is the vector of response variables, $\boldsymbol{X} = (\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n)^T$ is a design matrix, and

$$f(y_i \mid \boldsymbol{x}_i; \beta_0, \boldsymbol{\beta}) = \frac{(e^{\beta_0 + \boldsymbol{x}_i^T \boldsymbol{\beta}})^{y_i}}{1 + e^{\beta_0 + \boldsymbol{x}_i^T \boldsymbol{\beta}}}. \tag{2.3}$$

With the likelihood function of Eq. (2.2), the log-likelihood function is given by

$$\log f(\boldsymbol{y}|\boldsymbol{X};\beta_0,\boldsymbol{\beta}) = \sum_{i=1}^{n} \left[y_i(\beta_0 + \boldsymbol{x}_i^T\boldsymbol{\beta}) - \log\left\{1 + \exp(\beta_0 + \boldsymbol{x}_i^T\boldsymbol{\beta})\right\}\right]. \tag{2.4}$$

A maximum likelihood estimation is a method to estimate regression coefficients of the model by the maximization of a log-likelihood. However, maximum likelihood method often suffers from overfitting problems when the number of variables p is much larger than sample size n. Regularized likelihood methods solve this problem by imposing the penalty term on regression coefficients. Specifically, regression coefficients are estimated by solving the maximization problem as follows:

$$\max_{\boldsymbol{\beta}} \left\{\log f(\boldsymbol{y}|\boldsymbol{X};\beta_0,\boldsymbol{\beta}) - P_\lambda(\boldsymbol{\beta})\right\}, \tag{2.5}$$

where $P_\lambda(\boldsymbol{\beta})$ (> 0) is a penalty term with a regularization parameter λ (> 0), which adjusts the strength of the penalty. The larger the value of λ, the stronger an effect of regularization becomes. When an overfitting occurs, absolute values of regression coefficients tend to become too large. A penalty term prevents this problem and leads to a stable estimation. There are various types of penalty terms. For example, the regularization method which imposes the sum of square of regression coefficients (L_2 norm) as a penalty term is called ridge (Hoerl and Kennard, 1970). Regression coefficients of logistic regression model with L_2-norm penalty is given by solving the following maximization problem:

$$\max_{\boldsymbol{\beta}} \left\{ \log f(\boldsymbol{y}|\boldsymbol{X};\beta_0,\boldsymbol{\beta}) - \lambda \sum_{j=1}^{p} \beta_j^2 \right\}. \tag{2.6}$$

In ridge regression models, regression coefficients are shrunk toward zero and overfitting is prevented. However, ridge regression does not estimate regression coefficients as exactly zero. Therefore, the model will not become sparse and the benefits of sparsity such as variable selection are not brought.

2.2 Lasso

Lasso (Tibshirani, 1996) is a regularization method which induces the sparse estimators of regression coefficients. Lasso can be considered as the optimization problem of Eq. (2.5) where $P_\lambda(\boldsymbol{\beta}) = \lambda \sum_{i=1}^{p} |\beta_j|$. Lasso can be formulated as

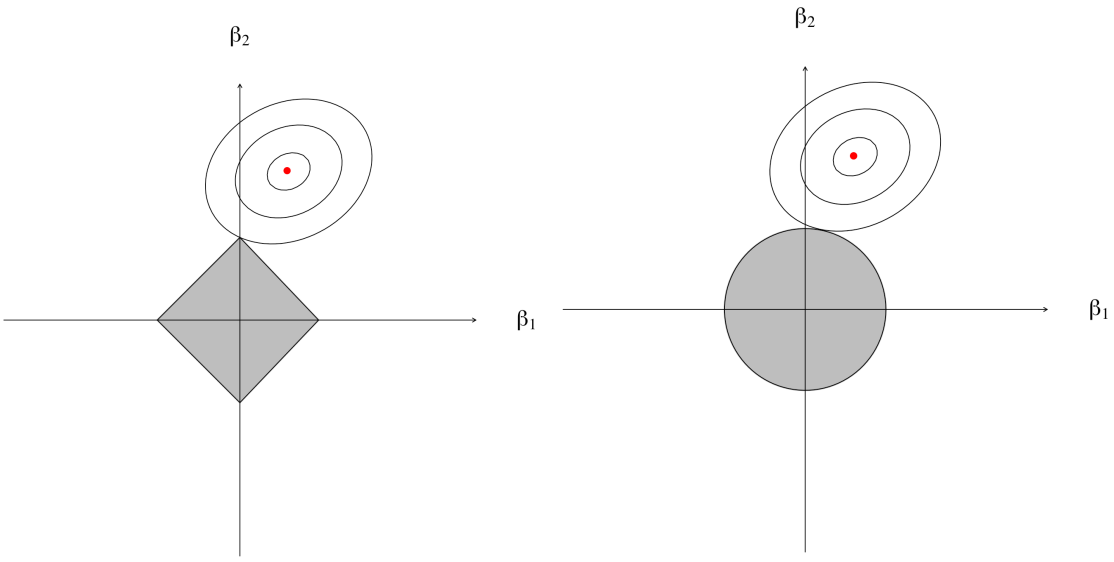
$$\max_{\boldsymbol{\beta}} \left\{ \log f(\boldsymbol{y}|\boldsymbol{X}; \beta_0, \boldsymbol{\beta}) - \lambda \sum_{j=1}^{p} |\beta_j| \right\}, \tag{2.7}$$

where λ (> 0) is the parameter which adjusts the degree of sparsity. When λ is large enough, lasso estimates some of the regression coefficients as exactly zero. The Eq. (2.7) can be rewritten as the constrained optimization problem:

$$\max_{\boldsymbol{\beta}} \log f(\boldsymbol{y}|\boldsymbol{X}; \beta_0, \boldsymbol{\beta}) \qquad \text{subject to} \quad \sum_{j=1}^{p} |\beta_j| \leq s_1, \tag{2.8}$$

while the constraint of ridge is $\sum_{j=1}^{p} \beta_j^2 \leq s_2$. Here, s_1 and s_2 are positive parameters adjusting the strength of the constraint. Lasso induces a sparsity because the constraint of lasso is the sum of absolute values of regression coefficients. Fig. 1 shows the constraint regions of lasso and ridge when $p = 2$. The solution of the problem in Eq. (2.8) is found at the intersection of the contours of log-likelihood and the constraint region. Unlike ridge, the constraint region of lasso has corners on the axes and the solution tends to be at those corners. Therefore, the regression coefficients of lasso tend to be estimated as exactly zero.

As some of regression coefficients become exactly zero, the corresponding variables are omitted from the model. For this feature, variable selection can be performed.



(a) lasso (b) ridge

Fig. 1: The constraint region of lasso and ridge. The region colored in gray in the figure (a) means the constraint region of lasso and the figure (b) means the one of ridge. The contour is the likelihood function and the red point is the maximum likelihood estimator.

2.3 Fused lasso

Fused lasso (Tibshirani et al., 2005) is an extension of lasso to analyze data with group structure. Specifically, fused lasso can be used when successive variables have similar roles for prediction. This situation often appears in real data analysis. For example, the successive pixels in an image data tend to have the same value. Fused lasso is formulated as the following optimization problem:

$$\max_{\boldsymbol{\beta}} \left\{ \log f(\boldsymbol{y}|\boldsymbol{X}; \beta_0, \boldsymbol{\beta}) - \lambda_1 \sum_{j=1}^{p} |\beta_j| - \lambda_2 \sum_{j=2}^{p} |\beta_j - \beta_{j-1}| \right\}, \tag{2.9}$$

where λ_1 (> 0) and λ_2 (> 0) are regularization parameters. λ_1 adjusts the degree of sparsity and λ_2 adjusts the smoothness of the successive differences of regression coefficients. Fused lasso in Eq. (2.9) can be rewritten as the constraint problem as

$$\max_{\boldsymbol{\beta}} \log f(\boldsymbol{y}|\boldsymbol{X}; \beta_0, \boldsymbol{\beta}) \qquad \text{subject to} \sum_{j=1}^{p} |\beta_j| \leq s_1 \text{ and} \sum_{j=2}^{p} |\beta_j - \beta_{j-1}| \leq s_2. \tag{2.10}$$

Figure 2 shows the constraint region of fused lasso when $\lambda_1 = \lambda_2$ and $p = 2$. The constraint region has the corners not only on the axes, but also on the points such that $\beta_1 = \beta_2$. The solution of Eq. (2.10) tends to be on these points and fused lasso estimates not only some of regression coefficients but also the successive differences of them as exactly zero.

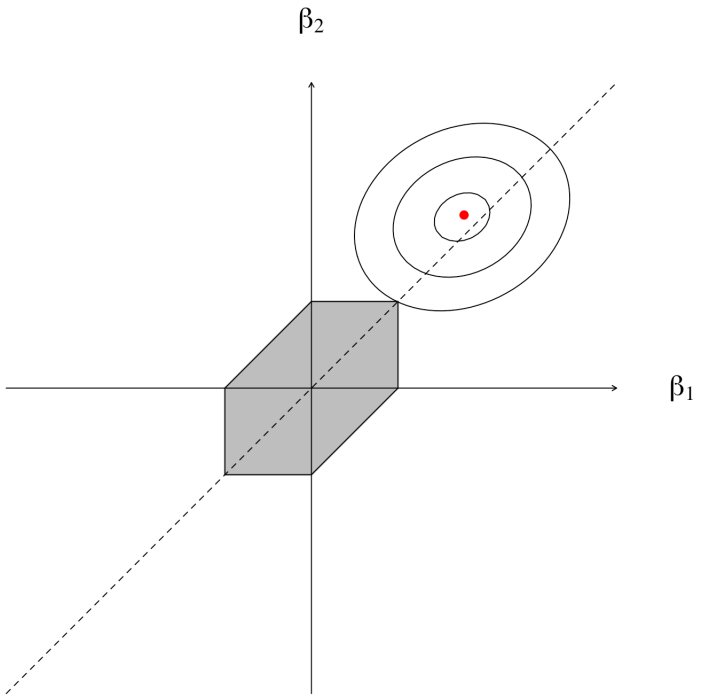


Fig. 2: The constraint region of lasso and ridge. The region colored in gray means the constraint region of fused lasso. The contour is the likelihood function and the red point is the maximum likelihood estimator.

For this property of the penalty, fused lasso has features of variable selection and variable fusion. Fused lasso estimates the regression coefficients of successive variables as the same by variable fusion. Therefore, fused lasso can be applied to data whose variables are some sense ordered. For a linear regression, Friedman et al. (2007) applied fused lasso to an image smoothing. In addition, a logistic regression model with fused lasso is used for a classification of spectral data (Yu et al., 2015).

3 Bayesian sparse logistic regression

As described in the former section, many frequentist regularization methods for sparse modeling are proposed. In Bayesian methods, regularization can be considered as assuming priors, which shrink the targets toward zero, on regression coefficients. In this section, we first introduce the Bayesian logistic regression model and the Bayesian parameter estimation methods. Next, we describe the Bayesian shrinkage methods with a Laplace prior. Then, we move on to the introduction of horseshoe prior and describe Bayesian lasso with horseshoe prior for binary data.

3.1 Bayesian estimation

In Bayesian approach, the parameters of the models are considered as a random variables. The parameters are estimated from the posterior distribution which can be obtained using Bayes' theorem. Specifically, Bayesian methods usually aim to obtain samples from the posterior distribution. For estimation procedure, Markov chain Monte Carlo (MCMC) methods are often utilized.

3.1.1 Bayesian logistic regression

Let us begin with logistic regression model with the regression coefficients $\boldsymbol{\beta}$ with the prior

$$\pi(\boldsymbol{\beta}) = \prod_{j=1}^{p} \mathrm{N}\left(\beta_j \mid 0, \frac{1}{2\lambda_1}\right), \tag{3.1}$$

where λ_1 is a positive parameter. Then, the logistic regression model is given by

$$\begin{aligned} y_i \mid \boldsymbol{x}_i, \boldsymbol{\beta}, \beta_0 &\sim \mathrm{Binom}\left(1, \frac{1}{1+e^{-(\beta_0+\boldsymbol{x}_i^T\boldsymbol{\beta})}}\right), \\ \beta_j \mid \lambda_1 &\sim \mathrm{N}\left(0, \frac{1}{2\lambda_1}\right). \end{aligned} \tag{3.2}$$

The posterior distribution of $\boldsymbol{\beta}$ is given by Bayes' theorem as

$$\pi(\boldsymbol{\beta}|\boldsymbol{X}, \boldsymbol{y}, \beta_0) = \frac{f(\boldsymbol{y}|\boldsymbol{X}; \beta_0, \boldsymbol{\beta})\pi(\boldsymbol{\beta})}{\int f(\boldsymbol{y}|\boldsymbol{X}; \beta_0, \boldsymbol{\beta})\pi(\boldsymbol{\beta})d\boldsymbol{\beta}}. \tag{3.3}$$

The denominator of Eq. (3.3) is called normalizing constant, which does not depend on $\boldsymbol{\beta}$. Hence, the posterior distribution can be expressed as

$$\pi(\boldsymbol{\beta}|\boldsymbol{X}, \boldsymbol{y}, \beta_0) \propto f(\boldsymbol{y}|\boldsymbol{X}; \beta_0, \boldsymbol{\beta})\pi(\boldsymbol{\beta}). \tag{3.4}$$

With the likelihood in Eq. (2.2), the posterior distribution of $\boldsymbol{\beta}$ can be calculated as follows:

$$\begin{aligned}
\pi(\boldsymbol{\beta}|\boldsymbol{X}, \boldsymbol{y}, \beta_0) &\propto f(\boldsymbol{y}|\boldsymbol{X}; \beta_0, \boldsymbol{\beta}) \prod_{j=1}^{p} \mathrm{N}\left(\beta_j \mid 0, \frac{1}{2\lambda_1}\right) \\
&\propto \prod_{i=1}^{n} f(y_i|\boldsymbol{x}_i; \beta_0, \boldsymbol{\beta}) \prod_{j=1}^{p} \mathrm{N}\left(\beta_j \mid 0, \frac{1}{2\lambda_1}\right) \\
&\propto \prod_{i=1}^{n} \frac{(e^{\beta_0 + \boldsymbol{x}_i^T\boldsymbol{\beta}})^{y_i}}{1 + e^{\beta_0 + \boldsymbol{x}_i^T\boldsymbol{\beta}}} \prod_{j=1}^{p} (\pi {\lambda_1}^{-1})^{-\frac{1}{2}} \exp\left\{-\lambda_1 \beta_j^2\right\}.
\end{aligned} \tag{3.5}$$

Then, the log posterior distribution is expressed as follows:

$$\log \pi(\boldsymbol{\beta}|\boldsymbol{X}, \boldsymbol{y}, \beta_0) \propto \sum_{i=1}^{n} \left[y_i(\beta_0 + \boldsymbol{x}_i^T\boldsymbol{\beta}) - \log\left\{1 + \exp(\beta_0 + \boldsymbol{x}_i^T\boldsymbol{\beta})\right\}\right] - \lambda_1 \sum_{j=1}^{p} \beta_j^2. \tag{3.6}$$

The maximization problem of Eq. (3.6) is same as the ridge regression expressed by the optimization problem in Eq. (2.6). The estimation of the parameter by the maximization of the posterior distribution as the above is called the maximum a posteriori (MAP) estimation.

3.1.2 Metropolis-Hastings algorithm

To estimate the parameters of the Bayesian model, MCMC methods are commonly used. The calculation of normalizing constant often cannot be analytically

and getting samples from the posterior distribution becomes difficult. MCMC methods are strong tools which enable a sampling from a complicated multivariate posterior distribution. The one of the most popular MCMC methods is Metropolis-Hastings (M-H) algorithm (Metropolis et al. (1953); Hastings (1970)). To get samples from probability distribution $\pi(\boldsymbol{\theta})$ (target distribution), $q(\boldsymbol{\theta} \mid \boldsymbol{\eta})$ (proposal distribution) is utilized in M-H algorithm. In comparison with the target distribution, the proposal distribution is often easier to sample from. $q(\boldsymbol{\theta} \mid \boldsymbol{\eta})$ is a probability distribution for next sample $\boldsymbol{\theta}$ given current sample $\boldsymbol{\eta}$. Specifically, M-H algorithm is constructed by Algorithm 1.

Algorithm 1 Metropolis-Hastings algorithm

Step 1: Initialize $\boldsymbol{\theta}_{(0)}$.

Step 2: Repeat (a) to (c) for $i = 1, 2, \ldots$.

(a) Generate $\boldsymbol{\eta}$ from $q(\boldsymbol{\eta} \mid \boldsymbol{\theta}_{(i)})$.

(b) Generate c from $\mathrm{U}(0, 1)$.

(c) Set $\boldsymbol{\theta}_{(i+1)}$ as

$$\boldsymbol{\theta}_{(i+1)} = \begin{cases} \boldsymbol{\eta} & c \leq p(\boldsymbol{\theta}_{(i)}, \boldsymbol{\eta}) \\ \boldsymbol{\theta}_{(i)} & \text{otherwise} \end{cases},$$

where

$$p(\boldsymbol{\theta}_{(i)}, \boldsymbol{\eta}) = \min\left\{1, \frac{\pi(\boldsymbol{\eta}) q(\boldsymbol{\theta}_{(i)} \mid \boldsymbol{\eta})}{\pi(\boldsymbol{\theta}_{(i)}) q(\boldsymbol{\eta} \mid \boldsymbol{\theta}_{(i)})}\right\}.$$

In Algorithm 1, $\mathrm{U}(\alpha, \beta)$ is a uniform distribution with hyper-parameters α and β. The probability density function of a uniform distribution is given by

$$\mathrm{U}(\alpha, \beta) = \begin{cases} \dfrac{1}{\beta - \alpha} & (\alpha \leq x \leq \beta) \\ 0 & \text{otherwise} \end{cases}. \tag{3.7}$$

M-H algorithm repeats to decide the next sample value by accepting the candidate from the proposal distribution with the acceptance probability p and enables the

sampling from $\pi(\boldsymbol{\theta})$.

When $\boldsymbol{\theta}$ can be divided into n parts as $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \ldots, \boldsymbol{\theta}_n)$, multiple-block Metropolis-Hastings algorithm can be used. Multiple-block Metropolis-Hastings algorithm is expressed as Algorithm 2.

Algorithm 2 Multiple-block Metropolis-Hastings algorithm

Step 1: Initialize $\boldsymbol{\theta}_{(0)} = (\boldsymbol{\theta}_{1(0)}, \boldsymbol{\theta}_{2(0)}, \ldots, \boldsymbol{\theta}_{n(0)})$.

Step 2: Repeat the following steps for $i = 1, 2, \ldots$.

Repeat (a) to (c) for $j = 1, 2, \ldots$.

(a) Generate $\boldsymbol{\eta}$ from $q_j(\boldsymbol{\eta} \mid \boldsymbol{\theta}_{j(i)}, \boldsymbol{\theta}_{-j(i)})$.

(b) Generate c from $\mathrm{U}(0, 1)$.

(c) Set $\boldsymbol{\theta}_{j(i+1)}$ as

$$\boldsymbol{\theta}_{j(i+1)} = \begin{cases} \boldsymbol{\eta} & c \leq p_j(\boldsymbol{\theta}_{j(i)}, \boldsymbol{\eta}) \\ \boldsymbol{\theta}_{j(i)} & \text{otherwise} \end{cases},$$

where

$$p_j(\boldsymbol{\theta}_{j(i)}, \boldsymbol{\eta}) = \min\left\{1, \frac{\pi(\boldsymbol{\eta} \mid \boldsymbol{\theta}_{-j(i)}) q_j(\boldsymbol{\theta}_{j(i)} \mid \boldsymbol{\eta}, \boldsymbol{\theta}_{-j(i)})}{\pi(\boldsymbol{\theta}_{j(i)} \mid \boldsymbol{\theta}_{-j(i)}) q_j(\boldsymbol{\eta} \mid \boldsymbol{\theta}_{j(i)}, \boldsymbol{\theta}_{-j(i)})}\right\}.$$

In Algorithm 2, $\boldsymbol{\theta}_{-j}$ is $(\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_{j-1}, \boldsymbol{\theta}_{j+1}, \ldots, \boldsymbol{\theta}_n)$. By dividing parameters into blocks, multiple-block M-H algorithm can reduce the estimation time than the regular M-H algorithm.

However, it is difficult to estimate the parameters of the model in Eq. (3.2) by M-H methods. Although the calculation of normalizing constant becomes unnecessary, the likelihood function of logistic regression in Eq. (2.2) is complicated form. Therefore, finding the appropriate proposal distribution is difficult and sampling process is often inefficient even if there is some proposal distribution.

3.1.3 Gibbs sampling

Deciding the proposal distribution in M-H methods is often the painstaking problem. In some cases, M-H methods can be utilized without the proposal distribution. It is called Gibbs sampling (Geman and Geman, 1984). Suppose that we have an n-blocked parameter $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \ldots, \boldsymbol{\theta}_n)$. Gibbs sampling is expressed as Algorithm 3.

Algorithm 3 Gibbs sampling

Step 1: Initialize $\boldsymbol{\theta}_{(0)} = (\boldsymbol{\theta}_{1(0)}, \boldsymbol{\theta}_{2(0)}, \ldots, \boldsymbol{\theta}_{n(0)})$.

Step 2: Repeat (1) to (n) for $i = 1, 2, \ldots$.

(1) Sample $\boldsymbol{\theta}_{1(i+1)}$ from $\pi(\boldsymbol{\theta}_1 \mid \boldsymbol{\theta}_{2(i)}, \boldsymbol{\theta}_{3(i)}, \ldots, \boldsymbol{\theta}_{n(i)})$.

(2) Sample ${\boldsymbol{\theta}_2}^{(i+1)}$ from $\pi(\boldsymbol{\theta}_2 \mid \boldsymbol{\theta}_{1(i+1)}, \boldsymbol{\theta}_{3(i)}, \ldots, \boldsymbol{\theta}_{n(i)})$.

$\vdots$

(n) Sample ${\boldsymbol{\theta}_n}^{(i+1)}$ from $\pi(\boldsymbol{\theta}_n \mid \boldsymbol{\theta}_{1(i+1)}, \boldsymbol{\theta}_{2(i+1)}, \ldots, \boldsymbol{\theta}_{n-1(i+1)})$.

In Algorithm 3, $\pi(\boldsymbol{\theta}_k \mid \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_{k-1}, \boldsymbol{\theta}_{k+1}, \ldots, \boldsymbol{\theta}_{n-1})$ is called the full conditional distribution.

Gibbs sampling is the case of multiple-block M-H algorithm in Algorithm 2 with

$$q_j(\boldsymbol{\eta} \mid \boldsymbol{\theta}_{j(i)}, \boldsymbol{\theta}_{-j(i)}) = \pi(\boldsymbol{\eta} \mid \boldsymbol{\theta}_{-j(i)}). \tag{3.8}$$

Then, the acceptance probability in Algorithm 2 is

$$p_j(\boldsymbol{\theta}_{j(i)}, \boldsymbol{\eta}) = \min\left\{1, \frac{\pi(\boldsymbol{\eta} \mid \boldsymbol{\theta}_{-j(i)})\pi(\boldsymbol{\theta}_{j(i)} \mid \boldsymbol{\theta}_{-j(i)})}{\pi(\boldsymbol{\theta}_{j(i)} \mid \boldsymbol{\theta}_{-j(i)})\pi(\boldsymbol{\eta} \mid \boldsymbol{\theta}_{-j(i)})}\right\} = 1. \tag{3.9}$$

Therefore, the candidates from the proposal distribution are always accepted in Gibbs sampling. Hence, Gibbs sampling is more efficient than M-H algorithms.

Polson et al. (2013) enabled Gibbs sampling estimation for logistic regression model by using the hierarchical expression of likelihood (Polson et al., 2013) with

Pólya-Gamma distribution. The probability density function of Pólya-Gamma distribution with hyper-parameters a (> 0) and b is expressed as follows:

$$\mathrm{PG}(x|a,b) = \frac{1}{2\pi^2}\sum_{k=1}^{\infty}\frac{g_k}{(k-\frac{1}{2})^2+\frac{c^2}{(4\pi^2)}}, \tag{3.10}$$

where g_k is an independent random variable that assumes the gamma distribution $\mathrm{Ga}(a,1)$. Pólya-Gamma distribution in Eq. (3.10) can be expressed with $\mathrm{PG}(x \mid a,0)$ follows:

$$\mathrm{PG}(x|a,b) \propto \exp\left(-\frac{b^2x}{2}\right)\mathrm{PG}(x|a,0). \tag{3.11}$$

By using the relationship of (3.11), Polson et al. (2013) showed that the likelihood function in Eq. (2.3) can be expressed as a hierarchical expression with the latent variables w_i following Pólya-Gamma distribution $\mathrm{PG}(1,0)$ as follows:

$$\begin{aligned} f(y_i|\boldsymbol{x}_i,\boldsymbol{\beta},\beta_0) &= \frac{(e^{\beta_0+\boldsymbol{x}_i^T\boldsymbol{\beta}})^{y_i}}{1+e^{\beta_0+\boldsymbol{x}_i^T\boldsymbol{\beta}}} \\ &= 2^{-1}\exp\{\kappa_i(\beta_0+\boldsymbol{x}_i^T\boldsymbol{\beta})\}\int_0^{\infty}\exp\left\{-\frac{w_i(\beta_0+\boldsymbol{x}_i^T\boldsymbol{\beta})^2}{2}\right\}\mathrm{PG}(w_i \mid 1,0)dw_i, \end{aligned} \tag{3.12}$$

where $\kappa_i = y_i - \dfrac{1}{2}$. With this likelihood in Eq. 3.12, the model in Eq (3.2) can be expressed as

$$\begin{aligned} y_i|\boldsymbol{x}_i,\boldsymbol{\beta},\beta_0,w_i &\sim \mathrm{Binom}\left(1,\frac{1}{1+e^{-(\beta_0+\boldsymbol{x}_i^T\boldsymbol{\beta})}}\right), \\ w_i &\sim \mathrm{PG}(1,0), \\ \boldsymbol{\beta} \mid \lambda_1 &\sim \mathrm{N}_p(\mathbf{0}_p,\boldsymbol{B}), \end{aligned} \tag{3.13}$$

where $\boldsymbol{B} = \mathrm{diag}\left(\dfrac{1}{2\lambda_1},\dfrac{1}{2\lambda_1},\ldots,\dfrac{1}{2\lambda_1}\right)$ and $\mathbf{0}_p$ is a p-dimensional vector of zeros.

From the priors in Eq. (3.13), the full conditional distribution can be obtained

as follows:

$$\begin{aligned}
\boldsymbol{\beta} \mid \boldsymbol{y}, \boldsymbol{X}, \beta_0, \lambda_1, w_1, \ldots, w_n &\sim \mathrm{N}_p(\boldsymbol{A}^{-1}\boldsymbol{X}^T\boldsymbol{W}(\boldsymbol{z} - \beta_0\mathbf{1}), \boldsymbol{A}^{-1}), \\
\boldsymbol{A} &= \boldsymbol{X}^T\boldsymbol{W}\boldsymbol{X} + \boldsymbol{B}^{-1}, \\
w_i \mid \boldsymbol{x}_i, \boldsymbol{\beta}, \beta_0 &\sim \mathrm{PG}(1, \beta_0 + \boldsymbol{x}_i^T\boldsymbol{\beta}),
\end{aligned}$$

where $\boldsymbol{W} = \mathrm{diag}(w_1, \ldots, w_n)$, $\boldsymbol{z} = (\kappa_1/w_1, \kappa_2/w_2, \ldots, \kappa_n/w_n)^T$, and $\mathbf{1}$ is an n-dimensional vector with all elements are 1. From the full conditional distribution, Gibbs sampler can be constructed.

3.2 Bayesian lasso

Bayesian lasso (Tibshirani (1996); Park and Casella (2008)) is the Bayesian estimation method for lasso. In Bayesian lasso, Laplace distributions are assumed on regression coefficients $\boldsymbol{\beta}$ as

$$\begin{aligned}
\pi(\boldsymbol{\beta}) &= \prod_{j=1}^{p} \mathrm{Laplace}\left(\beta_j \mid 0, \frac{1}{\lambda_1}\right) \\
&= \prod_{j=1}^{p} \frac{\lambda_1}{2} \exp\left(-\lambda_1|\beta_j|\right).
\end{aligned} \tag{3.14}$$

The probability density function of a Laplace distribution is given by

$$\mathrm{Laplace}(x|\mu, b) = \frac{1}{2b} \exp\left(-\frac{|x-\mu|}{b}\right) \tag{3.15}$$

with hyper-parameters μ, b (> 0). Figure 3 shows the probability density function of Laplace prior.

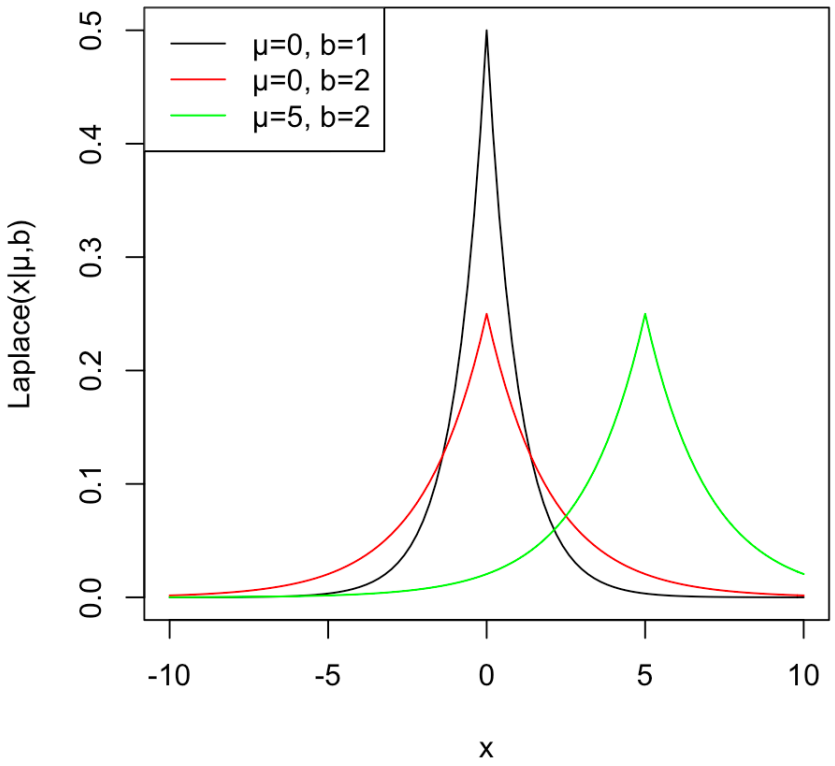


Fig. 3: The probability density function of Laplace distribution.

From Figure 3, Laplace distribution has the form which is high at zero and shrinks regression coefficients toward zero. The priors which shrink regression coefficients like Laplace prior are called shrinkage priors. Technically, Park and Casella (2008) proposed Bayesian lasso as a linear regression model. As Tibshirani (1996) mentioned, the lasso estimator for regression coefficients $\boldsymbol{\beta}$ is identical to a Bayesian posterior mode obtained by assuming prior as Eq. (3.14) on $\boldsymbol{\beta}$.

In the same way, lasso can be considered as a MAP estimation for a logistic regression model with Bayesian lasso. Logistic regression model with Bayesian lasso is expressed as follows:

$$\begin{aligned} y_i \mid \boldsymbol{x}_i, \boldsymbol{\beta}, \beta_0 &\sim \text{Binom}\left(1, \frac{1}{1+e^{-(\beta_0+\boldsymbol{x}_i^T\boldsymbol{\beta})}}\right), \\ \beta_j \mid \lambda_1 &\sim \text{Laplace}\left(0, \frac{1}{\lambda_1}\right). \end{aligned} \tag{3.16}$$

To consider a fully Bayesian estimation, we assume a uniform distribution $\mathrm{U}(-\alpha, \alpha)$ with hyper-parameter α on β_0. In addition, λ_1 is a positive parameter to adjust the degree of a shrinkage. As with Park and Casella (2008), we assume the gamma distribution as

$$\lambda_1^2 \sim \mathrm{Ga}(r_1, \delta_1), \tag{3.17}$$

where r_1 and δ_1 are positive hyper-parameters. The probability density function of the gamma distribution $\mathrm{Ga}(x|m, c),\ (m, c > 0)$ is given by:

$$\mathrm{Ga}(x|m, c) = \frac{c^m}{\Gamma(m)} x^{m-1} \exp(-cx) \quad (x \geq 0), \tag{3.18}$$

where $\Gamma(\cdot)$ is the gamma function.

In case we have given β_0 and λ_1, the conditional distribution of $\boldsymbol{\beta}$ follows:

$$\begin{aligned}
\pi(\boldsymbol{\beta}|\boldsymbol{X}, \boldsymbol{y}, \beta_0) &\propto f(\boldsymbol{y}|\boldsymbol{X}; \boldsymbol{\beta}, \beta_0) \prod_{j=1}^{p} \mathrm{Laplace}\left(\beta_j|0, \frac{1}{\lambda_1}\right) \\
&\propto \prod_{i=1}^{n} f(y_i|\boldsymbol{x}_i; \boldsymbol{\beta}, \beta_0) \prod_{j=1}^{p} \frac{\lambda_1}{2} \exp\left(-\lambda_1|\beta_j|\right) \\
&\propto \prod_{i=1}^{n} \frac{(e^{\beta_0 + \boldsymbol{x}_i^T\boldsymbol{\beta}})^{y_i}}{1 + e^{\beta_0 + \boldsymbol{x}_i^T\boldsymbol{\beta}}} \prod_{j=1}^{p} \frac{\lambda_1}{2} \exp\left(-\lambda_1|\beta_j|\right).
\end{aligned} \tag{3.19}$$

Then, the log posterior distribution is expressed as follows:

$$\log \pi(\boldsymbol{\beta}|\boldsymbol{X}, \boldsymbol{y}, \beta_0) \propto \sum_{i=1}^{n} \left[y_i(\beta_0 + \boldsymbol{x}_i^T\boldsymbol{\beta}) - \log\left\{1 + \exp(\beta_0 + \boldsymbol{x}_i^T\boldsymbol{\beta})\right\}\right] - \lambda_1 \sum_{j=1}^{p} |\beta_j|. \tag{3.20}$$

The maximization of Eq. (3.20) is the same form as lasso in Eq. (2.7). With the

likelihood in Eq. 3.12, the model in Eq (3.16) can be expressed as

$$\begin{aligned} y_i|\boldsymbol{x}_i, \boldsymbol{\beta}, \beta_0, w_i &\sim \text{Binom}\left(1, \frac{1}{1+e^{-(\beta_0+\boldsymbol{x}_i^T\boldsymbol{\beta})}}\right), \\ w_i &\sim \text{PG}(1,0), \\ \beta_j \mid \lambda_1 &\sim \text{Laplace}\left(0, \frac{1}{\lambda_1}\right). \end{aligned} \tag{3.21}$$

However, the easy estimation method still cannot be constructed because of the form of Laplace distribution. With a Laplace distribution, the posterior distribution is too complicated to get samples. Park and Casella (2008) proposed to use the hierarchical expression of Laplace distribution by using scale mixture of normals (Andrews and Mallows, 1974). With a scale mixture of normals, a Laplace distribution can be expressed hierarchically as follows:

$$\frac{\lambda}{2}\exp\left(-\lambda|\beta|\right) = \int_0^\infty \frac{1}{\sqrt{2\pi\tau^2}}\exp\left(-\frac{\beta^2}{2\tau^2}\right)\frac{\lambda^2}{2}\exp\left(-\frac{\lambda^2}{2}\tau^2\right)d\tau^2. \tag{3.22}$$

With the expression of Laplace distribution in Eq. (3.22), the prior in Eq. (3.14) can be expressed as

$$\begin{aligned} \pi(\boldsymbol{\beta}|\tau_1^2,\ldots,\tau_p^2) &= \prod_{j=1}^p \frac{1}{\sqrt{2\pi\tau_j^2}}\exp\left(-\frac{\beta_j^2}{2\tau_j^2}\right), \\ \pi(\tau_1^2,\ldots,\tau_p^2) &= \prod_{j=1}^p \frac{\lambda_1^2}{2}\exp\left(-\frac{\lambda_1^2}{2}\tau_j^2\right). \end{aligned}$$

Therefore, the logistic regression model in Eq. (3.16) can be rewritten in the form:

$$\begin{aligned} y_i|\boldsymbol{x}_i, \boldsymbol{\beta}, \beta_0 &\sim \text{Binom}\left(1, \frac{1}{1+e^{-(\beta_0+\boldsymbol{x}_i^T\boldsymbol{\beta})}}\right), \\ w_i &\sim \text{PG}(1,0), \\ \boldsymbol{\beta} \mid \tau_1^2,\ldots,\tau_p^2 &\sim \text{N}_p(\mathbf{0}_p, \boldsymbol{C}), \\ \tau_j^2 &\sim \text{EXP}\left(\frac{\lambda_1^2}{2}\right), \end{aligned} \tag{3.23}$$

where $\boldsymbol{C} = \text{diag}\left(\tau_1^2, \tau_2^2, \ldots, \tau_p^2\right)$. In addition, $\text{EXP}(x \mid d)$ is an exponential distri-

bution with parameter d (> 0) whose probability density function is given by

$$\mathrm{EXP}(x \mid d) = d \exp(-dx), \quad (x \geq 0). \tag{3.24}$$

From the likelihood in Eq. (3.12) and the priors in Eq. (3.23), the full conditional distributions can be obtained as:

$$\begin{aligned}
\boldsymbol{\beta} \mid \boldsymbol{y}, \boldsymbol{X}, \beta_0, \tau_1^2, \ldots, \tau_p^2, w_1, \ldots, w_n &\sim \mathrm{N}_p(\boldsymbol{A}^{-1}\boldsymbol{X}^T\boldsymbol{W}(\boldsymbol{z} - \beta_0\boldsymbol{1}), \boldsymbol{A}^{-1}), \\
\boldsymbol{A} &= \boldsymbol{X}^T\boldsymbol{W}\boldsymbol{X} + \boldsymbol{C}^{-1}, \\
w_i \mid \boldsymbol{x}_i, \boldsymbol{\beta}, \beta_0 &\sim \mathrm{PG}(1, \beta_0 + \boldsymbol{x}_i^T\boldsymbol{\beta}), \\
\frac{1}{\tau_j^2} \mid \beta_j, \lambda_1^2 &\sim \mathrm{IGauss}\left(\sqrt{\frac{\lambda_1^2}{\beta_j^2}}, \lambda_1^2\right), \\
\lambda_1^2 \mid \tau_1^2, \ldots, \tau_p^2 &\sim \mathrm{Ga}\left(p + r_1, \frac{1}{2}\sum_{j=1}^{p}\tau_j^2 + \delta_1\right), \\
\beta_0 \mid \boldsymbol{y}, \boldsymbol{X}, \tau_1^2, \ldots, \tau_p^2, w_1, \ldots, w_n &\sim \mathrm{N}\left(\frac{1}{S}\sum_{i=1}^{n} v_i, \frac{1}{S}\right), \\
\boldsymbol{v} &= (\boldsymbol{z} - \boldsymbol{X}\boldsymbol{\beta}) \odot \boldsymbol{W}, \\
S &= \sum_{i=1}^{n} w_i,
\end{aligned} \tag{3.25}$$

where $\odot$ means Hadamard product. In addition, $\mathrm{IGauss}(x|\mu, \lambda)$ is an inverse Gaussian distribution with the probability density function as follows:

$$\mathrm{IGauss}(x|\mu, \lambda) = \sqrt{\frac{\lambda}{2\pi}} x^{-\frac{3}{2}} \exp\left\{-\frac{\lambda(x-\mu)^2}{2\mu^2 x}\right\} \ (x > 0). \tag{3.26}$$

The Gibbs sampler can be constructed from the full conditional distribution in Eq. (3.25).

3.3 Bayesian fused lasso

Kyung et al. (2010) extended fused lasso as a Bayesian method called Bayesian fused lasso. Kyung et al. (2010) assumed Laplace distribution not only on regression coefficients but also on their successive differences. To apply the Bayesian fused lasso

method to logistic regression model, we can consider the following prior

$$\pi(\boldsymbol{\beta}) \propto \exp\left(-\lambda_1 \sum_{j=1}^{p} |\beta_j| - \lambda_2 \sum_{j=2}^{p} |\beta_j - \beta_{j-1}|\right). \tag{3.27}$$

By using scale mixture of normals (Andrews and Mallows, 1974), Eq. (3.27) can be expressed as follows:

$$\begin{aligned}\pi(\boldsymbol{\beta}) \propto & \prod_{j=1}^{p} \int \frac{1}{\sqrt{2\pi\tau_j^2}} \exp\left(-\frac{\beta_j^2}{2\tau_j^2}\right) \frac{\lambda_1^2}{2} \exp\left(-\frac{\lambda_1^2}{2}\tau_j^2\right) d\tau_j^2 \\ & \times \prod_{j=2}^{p} \int \frac{1}{\sqrt{2\pi\tilde{\tau}_j{}^2}} \exp\left\{-\frac{(\beta_j - \beta_{j-1})^2}{2\tilde{\tau}_j{}^2}\right\} \frac{\lambda_2^2}{2} \exp\left(-\frac{\lambda_2^2}{2}\tilde{\tau}_j{}^2\right) d\tilde{\tau}_j{}^2.\end{aligned} \tag{3.28}$$

As from the prior (3.28), the logistic regression model with Bayesian fused lasso can be given by

$$\begin{aligned} y_i \mid \boldsymbol{x}_i, \boldsymbol{\beta}, \beta_0, w_i &\sim \text{Binom}\left(1, \frac{1}{1 + e^{-(\beta_0 + \boldsymbol{x}_i^T \boldsymbol{\beta})}}\right), \\ w_i &\sim \text{PG}(1, 0), \\ \boldsymbol{\beta} \mid \tau_1^2, \ldots, \tau_p^2, \tilde{\tau}_2^2, \ldots, \tilde{\tau}_p^2 &\sim \text{N}_p(\mathbf{0}_p, \boldsymbol{D}), \\ \tau_j^2 &\sim \text{EXP}\left(\frac{\lambda_1^2}{2}\right), \\ \tilde{\tau}_j^2 &\sim \text{EXP}\left(\frac{\lambda_2^2}{2}\right), \end{aligned} \tag{3.29}$$

where the inverse matrix of $\boldsymbol{D}$ is expressed as

$$\boldsymbol{D}^{-1} = \begin{pmatrix} \frac{1}{\tau_1^2} + \frac{1}{\tilde{\tau}_2^2} & -\frac{1}{\tilde{\tau}_2^2} & 0 & \ldots & 0 & 0 \\ -\frac{1}{\tilde{\tau}_2^2} & \frac{1}{\tau_2^2} + \frac{1}{\tilde{\tau}_2^2} + \frac{1}{\tilde{\tau}_3^2} & -\frac{1}{\tilde{\tau}_3^2} & \ldots & 0 & 0 \\ 0 & -\frac{1}{\tilde{\tau}_3^2} & \frac{1}{\tau_3^2} + \frac{1}{\tilde{\tau}_3^2} + \frac{1}{\tilde{\tau}_4^2} & \ldots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \ldots & \frac{1}{\tau_{p-1}^2} + \frac{1}{\tilde{\tau}_{p-1}^2} + \frac{1}{\tilde{\tau}_p^2} & -\frac{1}{\tilde{\tau}_p^2} \\ 0 & 0 & 0 & \ldots & -\frac{1}{\tilde{\tau}_p^2} & \frac{1}{\tau_p^2} + \frac{1}{\tilde{\tau}_p^2} \end{pmatrix}.$$

As with Eq. (3.17) in Bayesian lasso, we assume the gamma distribution on λ_1

and λ_2 as

$$\begin{aligned} \lambda_1^2 &\sim \mathrm{Ga}(r_1, \delta_1), \\ \lambda_1^2 &\sim \mathrm{Ga}(r_2, \delta_2), \end{aligned} \tag{3.30}$$

where $r_1, \delta_1, r_2, \delta_2$ are positive hyper-parameters.

From the priors in Eq. (3.29), the full conditional distributions can be obtained as

$$\begin{aligned} \boldsymbol{\beta} \mid \boldsymbol{y}, \boldsymbol{X}, \beta_0, \tau_1^2, \ldots, \tau_p^2, \tilde{\tau}_2^2, \ldots, \tilde{\tau}_p^2, w_1, \ldots, w_n &\sim \mathrm{N}_p(\boldsymbol{A}^{-1}\boldsymbol{X}^T\boldsymbol{W}(\boldsymbol{z} - \beta_0\boldsymbol{1}), \boldsymbol{A}^{-1}), \\ \boldsymbol{A} &= \boldsymbol{X}^T\boldsymbol{W}\boldsymbol{X} + \boldsymbol{D}^{-1}, \\ w_i \mid \boldsymbol{x}_i, \boldsymbol{\beta}, \beta_0 &\sim \mathrm{PG}(1, \beta_0 + \boldsymbol{x}_i^T\boldsymbol{\beta}), \\ \frac{1}{\tau_j^2} \mid \beta_j, \lambda_1^2 &\sim \mathrm{IGauss}\left(\sqrt{\frac{\lambda_1^2}{\beta_j^2}}, \lambda_1^2\right), \\ \lambda_1^2 \mid \tau_1^2, \ldots, \tau_p^2 &\sim \mathrm{Ga}\left(p + r_1, \frac{1}{2}\sum_{j=1}^{p}\tau_j^2 + \delta_1\right), \\ \frac{1}{\tilde{\tau}_j^2} \mid \beta_j, \lambda_2^2 &\sim \mathrm{IGauss}\left(\sqrt{\frac{\lambda_2^2}{(\beta_j - \beta_{j-1})^2}}, \lambda_2^2\right), \\ \lambda_2^2 \mid \tilde{\tau}_2^2, \ldots, \tilde{\tau}_p^2 &\sim \mathrm{Ga}\left(p - 1 + r_2, \frac{1}{2}\sum_{j=1}^{p-1}\tilde{\tau}_j^2 + \delta_2\right), \\ \beta_0 \mid \boldsymbol{y}, \boldsymbol{X}, \boldsymbol{\beta}, \tau_1^2, \ldots, \tau_p^2, \tilde{\tau}_2^2, \ldots, \tilde{\tau}_p^2, w_1, \ldots, w_n &\sim \mathrm{N}\left(\frac{1}{S}\sum_{i=1}^{n} v_i, \frac{1}{S}\right), \\ \boldsymbol{v} &= (\boldsymbol{z} - \boldsymbol{X}\boldsymbol{\beta}) \odot \boldsymbol{W}, \\ S &= \sum_{i=1}^{n} w_i. \end{aligned} \tag{3.31}$$

From these full conditional distributions, the Gibbs sampler can be constructed.

3.4 Horseshoe prior

Bayesian lasso and Bayesian fused lasso are useful Bayesian sparse modeling methods. However, there is a problem caused by Laplace distribution. A Laplace distribution shrinks the regression coefficients too much. Therefore, it can be difficult to

fit the model with the structure of data. To overcome this problem, horseshoe prior is proposed by Carvalho et al. (2010).

The original horseshoe prior is considered with the p-dimensional vector as $\boldsymbol{y}$ which is $\boldsymbol{y} \mid \boldsymbol{\beta} \sim \mathrm{N}(\boldsymbol{\beta}, \sigma^2 \boldsymbol{I}_p)$. A horseshoe prior on $\boldsymbol{\beta}$ is defined as follows:

$$\beta_j|\lambda_j \sim \mathrm{N}(0, \lambda_j^2),\ \lambda_j|\tau \sim \mathrm{C}^+(0, \tau),\ \tau|\sigma \sim \mathrm{C}^+(0, \sigma), \tag{3.32}$$

where $\mathrm{C}^+(0, a)$ $(a > 0)$ is the half-Cauchy distribution which has a following probability density function:

$$p(x) = \frac{2a}{\pi\,(x^2 + a^2)}, \quad x > 0.$$

Carvalho et al. (2010) showed that the probability density function of a horseshoe prior with $\sigma^2 = \tau^2 = 1$ is given by

$$p(\theta) = \int_0^\infty \frac{1}{\sqrt{2\pi\lambda^2}} \exp\left(-\frac{\theta^2}{2\lambda^2}\right) \frac{2}{\pi(1+\lambda^2)} d\lambda. \tag{3.33}$$

Figure 4 shows the probability density function of horseshoe prior.

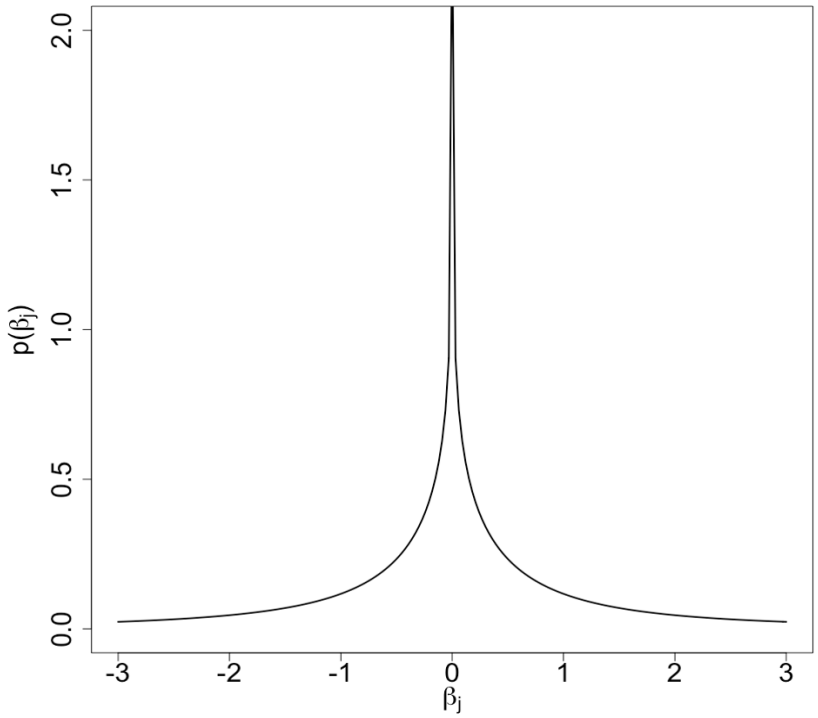


Fig. 4: The probability density function of horseshoe prior.

Horseshoe prior has both the infinite spike at zero and a Cauchy-like tail. The former shrinks the regression coefficients which are considered as small toward zero strongly and the latter prevents the large elements to be shrunk too much.

When $\sigma^2 = \tau^2 = 1$, the posterior mean of β_j given λ_j is

$$\pi(\beta_j \mid \lambda_j) \propto \exp\left\{ \frac{\left(\beta_j - \lambda_j^2 \kappa_j y_i\right)^2}{2\lambda_j^2 \kappa_j} \right\}, \tag{3.34}$$

where κ_j is given by

$$\kappa_j = \frac{1}{1+\lambda_j^2}. \tag{3.35}$$

Therefore, the posterior mean of β_j given λ_j is

$$\begin{aligned} \mathrm{E}(\beta_j \mid y_j, \lambda_j) &= \lambda_j^2 \kappa_j y_j, \\ &= (1-\kappa_j) y_j. \end{aligned} \tag{3.36}$$

Hence, the marginal posterior mean of β_j is

$$\begin{aligned} \mathrm{E}(\beta_j \mid y_j) &= \int_0^1 (1-\kappa_j) y_j p(\kappa_j \mid y_j) d\kappa_j \\ &\propto (1 - \mathrm{E}[\kappa_j \mid y_j]) y_j. \end{aligned} \tag{3.37}$$

κ_j is called a shrinkage factor, which indicates the degree of shrinkage of regression coefficients towards 0. When κ_j gets close to zero, the regression coefficient β_j is shrunk strongly towards 0. When κ_j is close to one, the regression coefficient β_j is not much shrunk. The probability density function of the shrinkage factor with $\sigma^2 = 1$ and $\tau^2 = 1$ is expressed as follows:

$$p(\kappa_j | \tau, \sigma) = \frac{1}{\pi} \frac{1}{\sqrt{\kappa_j}\sqrt{1-\kappa_j}}. \tag{3.38}$$

This is same as the one of the beta distribution $\mathrm{Beta}\left(x \mid \frac{1}{2}, \frac{1}{2}\right)$. The probability density function of the beta distribution is expressed as

$$\mathrm{Beta}(x \mid a, b) = \frac{x^{a-1}(1-x)^{b-1}}{B(a,b)}, \tag{3.39}$$

where a and b are positive hyper-parameters. Here, $B(\cdot)$ is a Beta function.

Figure 5 shows the probability density function of the shrinkage factor. From Figure 5, we can find that the form of the probability density function of the shrinkage factor looks like a horseshoe. The name of horseshoe prior comes from this shape.

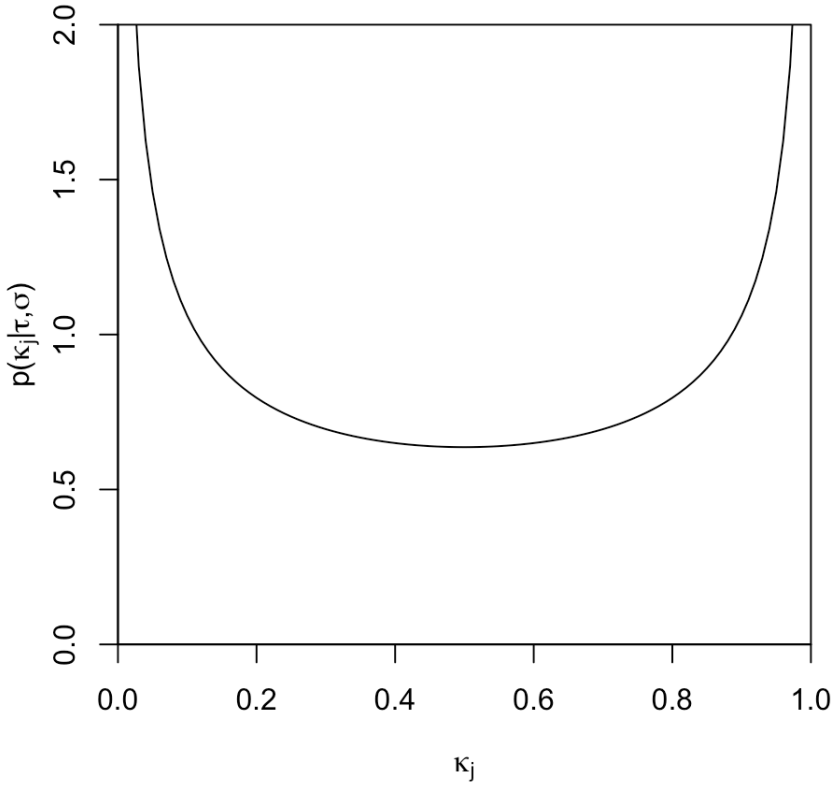


Fig. 5: The probability density function of shrinkage factor.

3.5 Bayesian logistic regression model with horseshoe prior

Makalic and Schmidt (2015) proposed the logistic regression model which imposes horseshoe prior on regression coefficients as follows:

$$\begin{aligned} y_i|\boldsymbol{x}_i, \boldsymbol{\beta}, \beta_0, w_i &\sim \text{Binom}\left(1, \frac{1}{1+e^{-(\beta_0+\boldsymbol{x}_i^T\boldsymbol{\beta})}}\right), \\ w_i &\sim \text{PG}(1,0), \\ \beta_j \mid \lambda_j^2, \tau^2 &\sim \text{N}_p(0, \lambda_j^2\tau^2), \\ \lambda_j &\sim \text{C}^+(0,1), \\ \tau &\sim \text{C}^+(0,1). \end{aligned} \tag{3.40}$$

Makalic and Schmidt (2015) enabled Gibbs sampling for the model in Eq. (3.40) with the hierarchical expression of the half-Cauchy distribution (Wand et al., 2011). Wand et al. (2011) showed that x follows $\mathrm{C}^{+}(0, A)$ when

$$x^2|a \sim \mathrm{IG}\left(\frac{1}{2}, \frac{1}{a}\right), \quad a \sim \mathrm{IG}\left(\frac{1}{2}, \frac{1}{A^2}\right), \tag{3.41}$$

where A is a positive constant. Here, $\mathrm{IG}(x|\nu_0, \eta_0)$ is the inverse-gamma distribution whose probability density function expressed as follows:

$$\mathrm{IG}(x|\nu_0, \eta_0) = \frac{\eta_0^{\nu_0}}{\Gamma(\nu_0)} x^{-\nu_0 - 1} \exp\left(-\frac{\eta_0}{x}\right) \quad (x > 0), \tag{3.42}$$

where ν_0, η_0 (> 0) are hyper-parameters.

Therefore, the half-Cauchy distributions $\mathrm{C}^{+}(0, 1)$ which λ_j and τ follow with the relationship of Eq. (3.41), Eq. (3.40) can be rewritten as

$$\begin{aligned}
y_i|\boldsymbol{x}_i, \boldsymbol{\beta}, \beta_0, w_i &\sim \mathrm{Binom}\left(1, \frac{1}{1 + e^{-(\beta_0 + \boldsymbol{x}_i^T \boldsymbol{\beta})}}\right), \\
w_i &\sim \mathrm{PG}(1, 0), \\
\boldsymbol{\beta} \mid \lambda_1^2, \ldots, \lambda_p^2, \tau^2 &\sim \mathrm{N}_p(\mathbf{0}_p, \boldsymbol{B}^{'}), \\
\lambda_j^2|\nu_j &\sim \mathrm{IG}\left(\frac{1}{2}, \frac{1}{\nu_j}\right), \\
\tau^2 \mid \xi &\sim \mathrm{IG}\left(\frac{1}{2}, \frac{1}{\xi}\right), \\
\nu_1, \ldots, \nu_p, \xi &\sim \mathrm{IG}\left(\frac{1}{2}, 1\right),
\end{aligned} \tag{3.43}$$

where $\boldsymbol{B}^{'}$ is $= \mathrm{diag}\left(\tau^2\lambda_1^2, \tau^2\lambda_2^2 \ldots, \tau^2\lambda_p^2\right)$.

Therefore, the full conditional distribution can be obtained as

$$\begin{aligned}
\boldsymbol{\beta} \mid \boldsymbol{y}, \boldsymbol{X}, \beta_0, \lambda_1^2, \ldots, \lambda_p^2, \tau, w_1, \ldots, w_n &\sim \mathrm{N}_p(\boldsymbol{A}^{-1}\boldsymbol{X}^T\boldsymbol{W}(\boldsymbol{z} - \beta_0\mathbf{1}), \boldsymbol{A}^{-1}), \\
\boldsymbol{A} &= \boldsymbol{X}^T\boldsymbol{W}\boldsymbol{X} + \boldsymbol{B}^{'-1}, \\
w_i \mid \boldsymbol{x}_i, \boldsymbol{\beta}, \beta_0 &\sim \mathrm{PG}(1, \beta_0 + \boldsymbol{x}_i^T\boldsymbol{\beta}), \\
\lambda_j^2 \mid \beta_j, \tau, \nu_j &\sim \mathrm{IG}\left(1, \frac{\beta_j^2}{2\tau^2} + \frac{1}{\nu_j}\right),
\end{aligned}$$

$$
\begin{aligned}
\tilde{\tau}^2 \mid \beta_1, \ldots, \beta_p, \lambda_1^2, \ldots, \lambda_p^2, \xi &\sim \mathrm{IG}\left(\frac{p+1}{2}, \frac{1}{2}\sum_{j=1}^{p}\frac{\beta_j^2}{\lambda_j^2} + \frac{1}{\xi}\right), \\
\nu_j \mid \lambda_j^2 &\sim \mathrm{IG}\left(1, \frac{1}{\lambda_j^2} + 1\right), \\
\xi \mid \tau^2 &\sim \mathrm{IG}\left(1, \frac{1}{\tau^2} + 1\right), \\
\beta_0 \mid \boldsymbol{y}, \boldsymbol{X}, \lambda_1^2, \ldots, \lambda_p^2, \tau^2, w_1, \ldots, w_n &\sim \mathrm{N}\left(\frac{1}{S}\sum_{i=1}^{n} v_i, \frac{1}{S}\right), \\
\boldsymbol{v} &= (\boldsymbol{z} - \boldsymbol{X}\boldsymbol{\beta}) \odot \boldsymbol{W}, \\
S &= \sum_{i=1}^{n} w_i.
\end{aligned}
$$

From these full conditional distributions, Gibbs sampler can be constructed.

4 Proposed method

The existing researches for logistic regression model utilize a Laplace prior to shrink the successive differences of regression coefficients. Assuming a Laplace prior on the differences of successive regression coefficients, those methods tend to lead to the over-shrinkage of differences which should not be zero. In this section, we propose a new logistic regression model which can remain large differences of successive regression coefficients by using horseshoe prior.

For logistic regression model in Eq. (2.1), we propose assuming a Laplace distribution on regression coefficients and the horseshoe prior on the successive differences of them as follows:

$$\begin{aligned}\pi(\boldsymbol{\beta}) \propto & \prod_{j=1}^{p} \text{Laplace}\left(\beta_j \,\middle|\, 0, \frac{1}{\tilde{\lambda}_1}\right) \\ & \times \int \left[\prod_{j=2}^{p} \int \frac{1}{\sqrt{2\pi\lambda_j^2\tilde{\tau}^2}} \exp\left\{-\frac{(\beta_j-\beta_{j-1})^2}{2\lambda_j^2\tilde{\tau}^2}\right\} \frac{2}{\pi(1+\lambda_j^2)} d\lambda_j^2\right] \frac{2}{\pi(1+\tilde{\tau}^2)} d\tilde{\tau}^2.\end{aligned} \tag{4.1}$$

By using the hierarchical expression of the log-likelihood with Pólya-Gamma distribution (Polson et al., 2013), the proposed model can be expressed as follows:

$$\begin{aligned} y_i \mid \boldsymbol{x}_i, \boldsymbol{\beta}, \beta_0, w_i &\sim \text{Binom}\left(1, \frac{1}{1+e^{-(\beta_0+\boldsymbol{x}_i^T\boldsymbol{\beta})}}\right), \\ w_i &\sim \text{PG}(1,0), \\ \beta_j \mid \tilde{\lambda}_1 &\sim \text{Laplace}\left(0, \frac{1}{\tilde{\lambda}_1}\right), \\ \beta_j - \beta_{j-1} | \lambda_j^2, \tilde{\tau}^2 &\sim \text{N}\left(0, \lambda_j^2\tilde{\tau}^2\right), \\ \lambda_j &\sim \text{C}^+(0,1), \\ \tilde{\tau} &\sim \text{C}^+(0,1). \end{aligned} \tag{4.2}$$

By assuming horseshoe prior on the successive differences of regression coefficients, the excess shrinkage of the differences which should not be zero is prevented.

By using scale mixture of normals (Andrews and Mallows, 1974) and the hierar-

chical expression of the half-Cauchy distribution (Wand et al., 2011), the prior (4.1) can be rewritten as

$$
\begin{aligned}
\pi(\boldsymbol{\beta}) \propto & \prod_{j=1}^{p} \int \frac{1}{\sqrt{2\pi\tau_j^2}} \exp\left(-\frac{\beta_j^2}{2\tau_j^2}\right) \frac{\tilde{\lambda}_1^2}{2} \exp\left(-\frac{\tilde{\lambda}_1^2}{2}\tau_j^2\right) d\tau_j^2 \\
& \times \int \left[\prod_{j=2}^{p} \int \frac{1}{\sqrt{2\pi\lambda_j^2\tilde{\tau}^2}} \exp\left\{-\frac{(\beta_j - \beta_{j-1})^2}{2\lambda_j^2\tilde{\tau}^2}\right\} \right. \\
& \left. \times \int \frac{(\frac{1}{\nu_j})^{\frac{1}{2}}}{\Gamma(\frac{1}{2})} (\lambda_j^2)^{-\frac{3}{2}} \exp\left(-\frac{1}{\nu_j\lambda_j^2}\right) \frac{1}{\Gamma(\frac{1}{2})} (\nu_j)^{-\frac{3}{2}} \exp\left(-\frac{1}{\nu_j}\right) d\nu_j d\lambda_j^2 \right] \\
& \times \int \frac{(\frac{1}{\xi})^{\frac{1}{2}}}{\Gamma(\frac{1}{2})} (\tilde{\tau}^2)^{-\frac{3}{2}} \exp\left(-\frac{1}{\xi\tilde{\tau}^2}\right) \frac{1}{\Gamma(\frac{1}{2})} (\xi)^{-\frac{3}{2}} \exp\left(-\frac{1}{\xi}\right) d\xi d\tilde{\tau}^2 \\
\propto & \int \dots \int \prod_{j=1}^{p} \frac{1}{\sqrt{2\pi\tau_j^2}} \exp\left(-\frac{\beta_j^2}{2\tau_j^2}\right) \prod_{j=1}^{p} \frac{\tilde{\lambda}_1^2}{2} \exp\left(-\frac{\tilde{\lambda}_1^2}{2}\tau_j^2\right) \\
& \times \prod_{j=2}^{p} \frac{1}{\sqrt{2\pi\lambda_j^2\tilde{\tau}^2}} \exp\left\{-\frac{(\beta_j - \beta_{j-1})^2}{2\lambda_j^2\tilde{\tau}^2}\right\} \prod_{j=2}^{p} \frac{(\frac{1}{\nu_j})^{\frac{1}{2}}}{\Gamma(\frac{1}{2})} (\lambda_j^2)^{-\frac{3}{2}} \exp\left(-\frac{1}{\nu_j\lambda_j^2}\right) \\
& \times \frac{(\frac{1}{\xi})^{\frac{1}{2}}}{\Gamma(\frac{1}{2})} (\tilde{\tau}^2)^{-\frac{3}{2}} \exp(-\frac{1}{\xi\tilde{\tau}^2}) \frac{1}{\Gamma(\frac{1}{2})} (\xi)^{-\frac{3}{2}} \exp\left(-\frac{1}{\xi}\right) \prod_{j=2}^{p} \frac{1}{\Gamma(\frac{1}{2})} (\nu_j)^{-\frac{3}{2}} \exp\left(-\frac{1}{\nu_j}\right) \\
& \times d\tilde{\tau}^2 d\xi \prod_{j=1}^{p} d\tau_j^2 \prod_{j=2}^{p} d\lambda_j^2 \prod_{j=2}^{p} d\nu_j \\
\propto & \int \dots \int (\tilde{\tau}^2)^{-\frac{p-1}{2}} \pi(\tilde{\tau}^2 \mid \xi) \pi(\xi) \prod_{j=1}^{p} (\tau_j^2)^{-\frac{1}{2}} \prod_{j=2}^{p} (\lambda_j^2)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}\boldsymbol{\beta}^T \boldsymbol{E}^{-1} \boldsymbol{\beta}\right) \\
& \times \prod_{j=1}^{p} \pi(\tau_j^2) \prod_{j=2}^{p} \pi(\lambda_j^2 \mid \nu_j) \prod_{j=2}^{p} \pi(\nu_j) d\tilde{\tau}^2 d\xi \prod_{j=1}^{p} d\tau_j^2 \prod_{j=2}^{p} d\lambda_j^2 \prod_{j=2}^{p} d\nu_j.
\end{aligned}
$$

Therefore, the model in Eq. (4.2) can be rewritten as follows:

$$
\begin{aligned}
y_i \mid \boldsymbol{x}_i, \boldsymbol{\beta}, \beta_0, w_i &\sim \text{Binom}\left(1, \frac{1}{1+e^{-(\beta_0+\boldsymbol{x}_i^T\boldsymbol{\beta})}}\right), \\
w_i &\sim \text{PG}(1,0), \\
\boldsymbol{\beta} \mid \tau_1^2, \ldots, \tau_p^2, \lambda_2^2, \ldots, \lambda_p^2, \tilde{\tau}^2 &\sim \text{N}_p(\mathbf{0}_p, \boldsymbol{E}), \\
\tau_j^2 &\sim \text{EXP}\left(\frac{\tilde{\lambda}_1^2}{2}\right), \\
\tilde{\tau}^2 \mid \xi &\sim \text{IG}\left(\frac{1}{2}, \frac{1}{\xi}\right), \\
\lambda_j^2 \mid \nu_j &\sim \text{IG}\left(\frac{1}{2}, \frac{1}{\nu_j}\right), \\
\xi, \nu_j &\sim \text{IG}\left(\frac{1}{2}, 1\right),
\end{aligned}
\tag{4.3}
$$

where the inverse matrix of $\boldsymbol{E}$ is $\boldsymbol{E}^{-1}$ expressed as

$$
\boldsymbol{E}^{-1} = \begin{pmatrix}
\frac{1}{\tau_1^2} + \frac{1}{\lambda_2^2\tilde{\tau}^2} & -\frac{1}{\lambda_2^2\tilde{\tau}^2} & 0 & \ldots & 0 & 0 \\
-\frac{1}{\lambda_2^2\tilde{\tau}^2} & \frac{1}{\tau_2^2} + \frac{1}{\lambda_2^2\tilde{\tau}^2} + \frac{1}{\lambda_3^2\tilde{\tau}^2} & -\frac{1}{\lambda_3^2\tilde{\tau}^2} & \ldots & 0 & 0 \\
0 & -\frac{1}{\lambda_3^2\tilde{\tau}^2} & \frac{1}{\tau_3^2} + \frac{1}{\lambda_3^2\tilde{\tau}^2} + \frac{1}{\lambda_4^2\tilde{\tau}^2} & \ldots & 0 & 0 \\
\vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\
0 & 0 & 0 & \ldots & \frac{1}{\tau_{p-1}^2} + \frac{1}{\lambda_{p-1}^2\tilde{\tau}^2} + \frac{1}{\lambda_p^2\tilde{\tau}^2} & -\frac{1}{\lambda_p^2\tilde{\tau}^2} \\
0 & 0 & 0 & \ldots & -\frac{1}{\lambda_p^2\tilde{\tau}^2} & \frac{1}{\tau_p^2} + \frac{1}{\lambda_p^2\tilde{\tau}^2}
\end{pmatrix}.
\tag{4.4}
$$

From the priors in Eq. (4.3), we can get the full conditional distributions as follows:

$$
\begin{aligned}
\boldsymbol{\beta} \mid \boldsymbol{y}, \boldsymbol{X}, \beta_0, \tau_1^2, \ldots, \tau_p^2, \lambda_2^2, \ldots, \lambda_p^2, \tilde{\tau}^2, w_1, \ldots, w_n &\sim \text{N}_p(\boldsymbol{A}^{-1}\boldsymbol{X}^T\boldsymbol{W}(\boldsymbol{z} - \beta_0\mathbf{1}), \boldsymbol{A}^{-1}), \\
\boldsymbol{A} &= \boldsymbol{X}^T\boldsymbol{W}\boldsymbol{X} + \boldsymbol{E}^{-1}, \\
w_i \mid \boldsymbol{x}_i, \boldsymbol{\beta}, \beta_0 &\sim \text{PG}(1, \beta_0 + \boldsymbol{x}_i^T\boldsymbol{\beta}), \\
\frac{1}{\tau_j^2} \mid \beta_j, \tilde{\lambda}_1^2 &\sim \text{IGauss}\left(\sqrt{\frac{\tilde{\lambda}_1^2}{\beta_j^2}}, \tilde{\lambda}_1^2\right), \\
\tilde{\lambda}_1^2 \mid \tau_1^2, \ldots, \tau_p^2 &\sim \text{Ga}\left(p + r_1, \frac{1}{2}\sum_{j=1}^{p}\tau_j^2 + \delta_1\right),
\end{aligned}
$$

$$
\begin{aligned}
\tilde{\tau}^2 \mid \beta_1, \ldots, \beta_p, \lambda_2^2, \ldots, \lambda_p^2, \xi &\sim \mathrm{IG}\left(\frac{p}{2}, \frac{1}{2}\sum_{j=2}^{p}\frac{(\beta_j - \beta_{j-1})^2}{\lambda_j^2} + \frac{1}{\xi}\right), \\
\lambda_j^2 \mid \beta_j, \beta_{j-1}, \tilde{\tau}^2, \nu_j &\sim \mathrm{IG}\left(1, \frac{(\beta_j - \beta_{j-1})^2}{2\tilde{\tau}^2} + \frac{1}{\nu_j}\right), \\
\nu_j \mid \lambda_j^2 &\sim \mathrm{IG}\left(1, \frac{1}{\lambda_j^2} + 1\right), \\
\xi \mid \tilde{\tau}^2 &\sim \mathrm{IG}\left(1, \frac{1}{\tilde{\tau}^2} + 1\right), \\
\beta_0 \mid \boldsymbol{y}, \boldsymbol{X}, \tau_1^2, \ldots, \tau_p^2, \lambda_2^2, \ldots, \lambda_p^2, \tilde{\tau}^2, w_1, \ldots, w_n &\sim \mathrm{N}\left(\frac{1}{S}\sum_{i=1}^{n} v_i, \frac{1}{S}\right), \\
v_i &= (\boldsymbol{z} - \boldsymbol{X}\boldsymbol{\beta}) \odot \boldsymbol{W}, \\
S &= \sum_{i=1}^{n} w_i.
\end{aligned}
$$

From the results above, we can construct the Gibbs sampler for the proposed model. The proposed method is also able to be applied to a linear regression model (Kakikawa et al., 2022). The detail is described in Appendix A.

5 Numerical studies

In this section, we compare the performance of the proposed method with the existing methods through the Monte Carlo Simulation.

We simulated artificial data from the true model:

$$\Pr(y_i = 1 \mid \boldsymbol{x}_i) = \frac{1}{1 + e^{-(\beta_0^* + \boldsymbol{x}_i^T \boldsymbol{\beta}^*)}}, \tag{5.1}$$

where β_0^* is a true intercept and $\boldsymbol{\beta}^*$ is a true p-dimensional regression coefficient vector. In addition, $\boldsymbol{x}_i$ $(i = 1, 2, \ldots, n)$ is multivariate normally distributed as $\mathrm{N}(\mathbf{0}_p, \boldsymbol{\Sigma})$.

We considered the following settings:

Case 1: $\boldsymbol{\beta}^* = (\mathbf{1.0}_5^T, \mathbf{0.0}_5^T, \mathbf{1.0}_5^T, \mathbf{0.0}_5^T)^T$, $\Sigma_{ii} = 1$, $\Sigma_{ij} = \rho$, $(i \neq j)$,
Case 2: $\boldsymbol{\beta}^* = (\mathbf{-1.0}_5^T, \mathbf{0.0}_5^T, \mathbf{1.0}_5^T, \mathbf{0.0}_5^T)^T$, $\Sigma_{ii} = 1$, $\Sigma_{ij} = \rho$, $(i \neq j)$,

where Σ_{ij} is the (i, j)-th element of $\boldsymbol{\Sigma}$. For Cases 1 and 2, we consider $\rho = 0, 0.5, 0.5^{|i-j|}, 0.7, 0.9$, and the sample sizes $n = 100, 300$. We simulated 100 datasets for each case.

We compared the logistic regression model with Bayesian fused lasso with horseshoe prior, which is the proposed method (LBFH), with the logistic regression model with fused lasso (LFL), and the logistic regression model with Bayesian fused lasso (LBFL). For LFL, we used the package `lqa` of the software **R**. The package can be obtained from `http://cran.nexr.com/web/packages/lqa/index.html`. The hyper-parameters λ_1 and λ_2 for LFL are selected by AIC (Akaike's Information Criterion). We used the number of non-zero regression coefficients for the term of the degrees of freedom in AIC. We set 15 candidates between 0.1 and 4.0 for each parameters. For LBFL and LBFH, the Gibbs sampling was run with 5,000 iterations (with the first 2,000 samples discarded as burn-in).

To assess the accuracy of estimation of regression coefficients, we computed mean

squared error (MSE):

$$\mathrm{MSE} = \frac{1}{100}\sum_{k=1}^{100}(\hat{\boldsymbol{\beta}}^{(k)} - \boldsymbol{\beta}^*)^T(\hat{\boldsymbol{\beta}}^{(k)} - \boldsymbol{\beta}^*), \tag{5.2}$$

where $\hat{\boldsymbol{\beta}}^{(k)} = (\hat{\beta}_1^{(k)}, \ldots, \hat{\beta}_p^{(k)})^T$ is the estimated values of regression coefficients for k-th data. In addition, we computed $\mathrm{MSE}_{\mathrm{diff}}$ to evaluate the accuracy of the estimation of the successive differences of regression coefficients. $\mathrm{MSE}_{\mathrm{diff}}$ is defined as

$$\mathrm{MSE}_{\mathrm{diff}} = \frac{1}{100}\sum_{k=1}^{100}\left[\sum_{j=2}^{p}\left\{(\hat{\beta}_j^{(k)} - \hat{\beta}_{j-1}^{(k)}) - (\beta_j^* - \beta_{j-1}^*)\right\}^2\right]. \tag{5.3}$$

To assess the prediction accuracy, we computed the negative expected log-likelihood (EL). We generated 1,000 test data and used the mean value of negative log-likelihood for them as EL. EL is defined as follows:

$$\mathrm{EL} = \frac{1}{100}\sum_{k=1}^{100}\left\{-\frac{1}{1000}\sum_{j=1}^{1000}\left[y_j^{\dagger(k)}(\boldsymbol{x}^{\dagger(k)T}_j\hat{\boldsymbol{\beta}}^{(k)} + \hat{\beta}_0^{(k)}) - \log\{1 + \exp(\hat{\boldsymbol{\beta}}^{(k)} + \hat{\beta}_0^{(k)})\}\right]\right\}, \tag{5.4}$$

where $y_j^{\dagger(k)}$ and $\boldsymbol{x}^{\dagger(k)}_j$ are the j-th data in the test data which are generated to evaluate the model estimated in the k-th dataset. $\hat{\beta}_0^{(k)}$ is the estimated value of the intercept for k-th dataset.

Table 1: MSE (standard deviation), $\mathrm{MSE}_{\mathrm{diff}}$, and EL for Case 1 and $\rho = 0$. Bold font indicates the smallest value among LFL, LBFL, and LBFH.

	$n = 100$			$n = 300$		
	MSE (sd)	$\mathrm{MSE}_{\mathrm{diff}}$ (sd)	EL (sd)	MSE (sd)	$\mathrm{MSE}_{\mathrm{diff}}$ (sd)	EL (sd)
LFL	**1.568** (0.782)	1.867 (1.270)	42.240 (3.555)	0.508 (0.236)	0.621 (0.371)	111.35 (3.061)
LBFL	2.565 (1.812)	3.351 (1.732)	46.178 (4.701)	0.719 (0.474)	1.054 (0.476)	114.24 (3.831)
LBFH	1.726 (0.773)	**1.300** (0.508)	**41.123** (2.427)	**0.376** (0.195)	**0.322** (0.212)	**109.09** (2.134)

Table 2: MSE (standard deviation), MSE_{diff}, and EL for Case 1 and $\rho = 0.5$. Bold font indicates the smallest value among LFL, LBFL, and LBFH.

	$n = 100$			$n = 300$		
	MSE (sd)	MSE_{diff} (sd)	EL (sd)	MSE (sd)	MSE_{diff} (sd)	EL (sd)
LFL	**3.266** (1.291)	2.894 (1.824)	**22.693** (2.656)	**1.525** (1.048)	1.827 (1.646)	**58.369** (3.616)
LBFL	4.507 (1.963)	5.345 (2.198)	25.029 (3.653)	2.195 (1.150)	3.276 (1.534)	61.501 (4.070)
LBFH	4.105 (0.912)	**2.616** (0.383)	22.773 (2.100)	1.703 (0.732)	**1.541** (0.522)	58.482 (2.879)

Table 3: MSE (standard deviation), MSE_{diff}, and EL for Case 1 and $\rho = 0.5^{|i-j|}$. Bold font indicates the smallest value among LFL, LBFL, and LBFH.

	$n = 100$			$n = 300$		
	MSE (sd)	MSE_{diff} (sd)	EL (sd)	MSE (sd)	MSE_{diff} (sd)	EL (sd)
LFL	**1.633** (0.839)	2.176 (1.561)	31.328 (3.405)	0.739 (0.455)	1.230 (0.986)	82.938 (3.266)
LBFL	3.270 (1.927)	4.856 (2.160)	34.972 (4.982)	1.249 (0.849)	2.327 (1.204)	85.797 (3.876)
LBFH	1.724 (0.708)	**1.693** (0.539)	**31.096** (2.697)	**0.557** (0.276)	**0.780** (0.439)	**81.692** (2.421)

Table 4: MSE (standard deviation), MSE_{diff}, and EL for Case 1 and $\rho = 0.7$. Bold font indicates the smallest value among LFL, LBFL, and LBFH.

	$n = 100$			$n = 300$		
	MSE (sd)	MSE_{diff} (sd)	EL (sd)	MSE (sd)	MSE_{diff} (sd)	EL (sd)
LFL	**4.241** (1.285)	3.174 (1.536)	19.601 (2.545)	**2.329** (1.386)	2.671 (2.118)	51.149 (3.242)
LBFL	5.213 (2.035)	5.385 (1.986)	21.293 (3.777)	3.239 (1.588)	4.607 (2.010)	53.410 (4.037)
LBFH	4.589 (0.635)	**2.783** (0.194)	**19.445** (2.198)	2.597 (0.885)	**2.058** (0.502)	**51.148** (2.659)

Table 5: MSE (standard deviation), MSE_{diff}, and EL for Case 1 and $\rho = 0.9$. Bold font indicates the smallest value among LFL, LBFL, and LBFH.

	$n = 100$			$n = 300$		
	MSE (sd)	MSE_{diff} (sd)	EL (sd)	MSE (sd)	MSE_{diff} (sd)	EL (sd)
LFL	5.418 (1.609)	3.366 (1.402)	16.995 (3.102)	**4.402** (1.964)	3.561 (2.226)	45.565 (3.038)
LBFL	5.548 (2.061)	4.664 (1.566)	17.565 (3.939)	4.692 (2.046)	5.521 (2.501)	46.059 (3.334)
LBFH	**5.041** (0.478)	**2.932** (0.072)	**16.787** (2.716)	4.068 (0.708)	**2.724** (0.218)	**45.179** (2.454)

Table 6: MSE (standard deviation), MSE_{diff}, and EL for Case 2 and $\rho = 0$. Bold font indicates the smallest value among LFL, LBFL, and LBFH.

	$n = 100$			$n = 300$		
	MSE (sd)	MSE_{diff} (sd)	EL (sd)	MSE (sd)	MSE_{diff} (sd)	EL (sd)
LFL	**1.635** (1.135)	1.914 (1.818)	42.629 (5.366)	0.509 (0.268)	0.655 (0.468)	112.08 (3.617)
LBFL	2.442 (1.512)	3.400 (1.910)	46.719 (6.028)	0.736 (0.349)	1.067 (0.498)	115.01 (3.694)
LBFH	1.645 (0.746)	**1.242** (0.569)	**41.193** (2.900)	**0.381** (0.244)	**0.324** (0.243)	**109.67** (2.768)

Table 7: MSE (standard deviation), MSE_{diff}, and EL for Case 2 and $\rho = 0.5$. Bold font indicates the smallest value among LFL, LBFL, and LBFH.

	$n = 100$			$n = 300$		
	MSE (sd)	MSE_{diff} (sd)	EL (sd)	MSE (sd)	MSE_{diff} (sd)	EL (sd)
LFL	**1.917** (1.709)	2.319 (2.536)	50.577 (4.808)	0.651 (0.374)	0.864 (0.597)	137.864 (3.571)
LBFL	2.487 (1.533)	3.756 (2.126)	53.336 (4.669)	0.885 (0.361)	1.429 (0.624)	140.53 (3.503)
LBFH	2.099 (0.918)	**1.602** (0.592)	**50.109** (2.448)	**0.541** (0.289)	**0.528** (0.300)	**136.35** (2.670)

Table 8: MSE (standard deviation), MSE_{diff}, and EL for Case 2 and $\rho = 0.5^{|i-j|}$. Bold font indicates the smallest value among LFL, LBFL, and LBFH.

	$n = 100$			$n = 300$		
	MSE (sd)	MSE_{diff} (sd)	EL (sd)	MSE (sd)	MSE_{diff} (sd)	EL (sd)
LFL	1.805 (1.058)	2.569 (2.586)	32.078 (3.566)	0.768 (0.662)	1.221 (1.123)	83.892 (3.885)
LBFL	2.640 (1.286)	4.300 (2.201)	34.666 (3.875)	1.286 (0.742)	2.331 (1.127)	87.082 (4.065)
LBFH	**1.768** (0.606)	**1.660** (0.467)	**31.380** (2.155)	**0.533** (0.272)	**0.722** (0.439)	**82.594** (2.793)

Table 9: MSE (standard deviation), MSE_{diff}, and EL for Case 2 and $\rho = 0.7$. Bold font indicates the smallest value among LFL, LBFL, and LBFH.

	$n = 100$			$n = 300$		
	MSE (sd)	MSE_{diff} (sd)	EL (sd)	MSE (sd)	MSE_{diff} (sd)	EL (sd)
LFL	**2.348** (1.937)	2.773 (2.996)	**56.085** (3.898)	0.817 (0.461)	1.150 (0.874)	155.636 (3.295)
LBFL	2.807 (1.465)	4.208 (2.272)	58.053 (3.528)	1.093 (0.420)	1.853 (0.869)	157.99 (3.129)
LBFH	6.859 (1.716)	**2.696** (0.233)	67.321 (1.834)	**0.742** (0.377)	**0.807** (0.446)	**154.76** (2.511)

Table 10: MSE (standard deviation), MSE_{diff}, and EL for Case 2 and $\rho = 0.9$. Bold font indicates the smallest value among LFL, LBFL, and LBFH.

	$n = 100$			$n = 300$		
	MSE (sd)	MSE_{diff} (sd)	EL (sd)	MSE (sd)	MSE_{diff} (sd)	EL (sd)
LFL	**3.930** (1.887)	3.848 (2.953)	**64.965** (2.001)	**1.622** (0.804)	2.002 (1.239)	**186.18** (2.560)
LBFL	3.957 (1.780)	4.716 (2.754)	65.415 (2.076)	2.095 (0.786)	3.110 (1.325)	187.87 (2.594)
LBFH	6.877 (1.727)	**2.695** (0.230)	67.341 (1.857)	1.943 (0.957)	**1.655** (0.582)	186.70 (2.499)

The results are summarized in Table 1 to Table 10. LBFH achieves the smallest EL in many cases. Although LFL gives the smallest MSE in seven situations in Case 1 and in five situations in Case 2, LBFH gives the smallest $\mathrm{MSE}_{\mathrm{diff}}$ in all cases. It might because LFL estimates the differences of successive non-zero regression coefficients largely but regression coefficients are estimated closely to the true ones. Let us give a simple example. For the true successive regression coefficients $\beta_1^* = 1.0$ and $\beta_2^* = 1.0$, LFL might tend to give the estimation value like $\hat{\beta}_1 = 1.5$ and $\hat{\beta}_2 = 0.5$. Hence, both $|\beta_1^* - \hat{\beta}_1|$ and $|\beta_2^* - \hat{\beta}_2|$ are 0.5. However, $|\hat{\beta}_2 - \hat{\beta}_1|$ becomes 1.0 while $|\beta_2^* - \beta_1^*|$ is 0.0. Therefore, $\mathrm{MSE}_{\mathrm{diff}}$ tends to become large, considering the value of MSE. When we compare Bayesian methods LBFL and LBFH, LBFH gives smaller MSE, $\mathrm{MSE}_{\mathrm{diff}}$ and EL in over 90% of all cases. On the whole, LBFH estimates the successive differences of regression coefficients more accurately and gives the better prediction performance than the existing methods. We note that LBFH does not have hyper-parameters which need to be selected by the analyst while LFL does. From this perspective, LBFH can be said to be more objective and effortless method than LFL.

6 Conclusion

In this paper, we proposed the logistic regression model with Bayesian fused lasso with horseshoe prior. By assuming a horseshoe prior on the successive differences of regression coefficients, the proposed model could prevent the over-shrinkage of relatively large successive differences. The results of numerical studies showed that our proposed method outperformed the existing methods in the estimation accuracy of the successive differences of regression coefficients and the prediction accuracy.

The proposed method does not estimate the successive differences of regression coefficients as exactly zero. Therefore, the exact blocks of regression coefficients are not produced. To find the exact groups of variables, the information of samples obtained by MCMC method can be utilized. In addition, the estimation time for our proposed model might be reduced by the sampling methods proposed in Johndrow et al. (2020) and Bhattacharya et al. (2016). We keep them as a future work.

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A Linear regression model with Bayesian fused lasso via horseshoe prior

Bayesian fused lasso modeling with horseshoe prior can be applied to linear regression model.

Suppose that we have an n-dimensional vector of the response variables $\boldsymbol{y} = (y_1, y_2, \ldots, y_n)^T$ and an $n \times p$ design matrix $\boldsymbol{X} = (\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n)^T$. Linear regression model is formulated as

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \tag{A.1}$$

where $\boldsymbol{\epsilon} = (\epsilon_1, \epsilon_2, \ldots, \epsilon_n)^T$ is an n-dimensional error vector which is distributed as $\mathrm{N}_n(\boldsymbol{0}_n, \sigma^2 I_n)$. Here, the response variable is supposed to be centered and the explanatory variable is supposed to be standardized as follows:

$$\sum_{i=1}^{n} y_i = 0, \quad \sum_{i=1}^{n} x_{ij} = 0, \quad \sum_{i=1}^{n} x_{ij}^2 = n \quad (j = 1, 2, \ldots, p). \tag{A.2}$$

The likelihood function is

$$f(\boldsymbol{y} \mid \boldsymbol{X}; \boldsymbol{\beta}, \sigma^2) = \prod_{i=1}^{n} f(y_i \mid \boldsymbol{x}_i; \boldsymbol{\beta}, \sigma^2), \tag{A.3}$$

where

$$f(y_i \mid \boldsymbol{x}_i; \boldsymbol{\beta}, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{ -\frac{(y_i - \boldsymbol{x}_i^T \boldsymbol{\beta})^2}{2\sigma^2} \right\}. \tag{A.4}$$

Kakikawa et al. (2022) proposed assuming the following prior on regression coefficients:

$$\begin{aligned} \pi(\boldsymbol{\beta} \mid \sigma^2) \propto (\sigma^2)^{-\frac{p}{2}} \prod_{j=1}^{p} \mathrm{Laplace}\left(\frac{\beta_j}{\sqrt{\sigma^2}} \,\middle|\, 0, \frac{1}{\tilde{\lambda}_1} \right) \\ \times \int \left[\prod_{j=2}^{p} \int \frac{1}{\sqrt{2\pi\lambda_j^2\tilde{\tau}^2\sigma^2}} \exp\left\{ -\frac{(\beta_j - \beta_{j-1})^2}{2\lambda_j^2\tilde{\tau}^2\sigma^2} \right\} \frac{2}{\pi(1+\lambda_j^2)} d\lambda_j^2 \right] \frac{2}{\pi(1+\tilde{\tau}^2)} d\tilde{\tau}^2. \end{aligned} \tag{A.5}$$

Here, the priors on σ^2, $\tilde{\lambda}_1^2$ are assumed as follows:

$$\begin{aligned} \sigma^2 &\sim \mathrm{IG}(\nu_0, \eta_0), \\ \tilde{\lambda}_1^2 &\sim \mathrm{Ga}(r_1, \delta_1). \end{aligned} \tag{A.6}$$

By using scale mixture of normals (Andrews and Mallows, 1974) and the hierarchical expression of the half-Cauchy distribution (Wand et al., 2011), the proposed model in Kakikawa et al. (2022) is expressed as follows:

$$\begin{aligned} \boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{\beta}, \sigma^2 &\sim \mathrm{N}_n(\boldsymbol{X}\boldsymbol{\beta}, \sigma^2 \boldsymbol{I}_n), \\ \boldsymbol{\beta} \mid \sigma^2, \tau_1^2, \ldots, \tau_p^2, \lambda_2^2, \ldots, \lambda_p^2, \tilde{\tau}^2 &\sim \mathrm{N}_p(\boldsymbol{0}_p, \sigma^2 \boldsymbol{E}), \\ \tau_j^2 &\sim \mathrm{EXP}\left(\frac{\tilde{\lambda}_1^2}{2}\right), \\ \tilde{\tau}^2 \mid \xi &\sim \mathrm{IG}\left(\frac{1}{2}, \frac{1}{\xi}\right), \\ \lambda_j^2 \mid \nu_j &\sim \mathrm{IG}\left(\frac{1}{2}, \frac{1}{\nu_j}\right), \\ \xi, \nu_j &\sim \mathrm{IG}\left(\frac{1}{2}, 1\right). \end{aligned}$$

As with the logistic regression model, the full conditional distributions can be obtained as

$$\begin{aligned} \boldsymbol{\beta} \mid \boldsymbol{y}, \boldsymbol{X}, \sigma^2, \tau_1^2, \ldots, \tau_p^2, \lambda_2^2, \ldots, \lambda_p^2, \tilde{\tau}^2 &\sim \mathrm{N}_p(\boldsymbol{A}^{-1}\boldsymbol{X}^T\boldsymbol{y}, \sigma^2 \boldsymbol{A}^{-1}), \\ \boldsymbol{A} &= \boldsymbol{X}^T\boldsymbol{X} + \boldsymbol{E}^{-1}, \\ \sigma^2 \mid \boldsymbol{y}, \boldsymbol{X}, \boldsymbol{\beta}, \tau_1^2, \ldots, \tau_p^2, \lambda_2^2, \ldots, \lambda_p^2, \tilde{\tau}^2 &\sim \mathrm{IG}\left(\frac{n_1}{2}, \frac{s_1}{2}\right), \\ n_1 &= n + 2p - 1 + \nu_0, \\ s_1 &= (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})^T(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) + \boldsymbol{\beta}^T\boldsymbol{E}^{-1}\boldsymbol{\beta} + \eta_0, \\ \frac{1}{\tau_j^2} \mid \beta_j, \sigma^2, \tilde{\lambda}_1^2 &\sim \mathrm{IGauss}\left(\sqrt{\frac{\sigma^2\tilde{\lambda}_1^2}{\beta_j^2}}, \tilde{\lambda}_1^2\right), \\ \tilde{\lambda}_1^2 \mid \tau_1^2, \ldots, \tau_p^2 &\sim \mathrm{Ga}\left(p + r_1, \frac{1}{2}\sum_{j=1}^{p}\tau_j^2 + \delta_1\right), \end{aligned}$$

$$\tilde{\tau}^2 \mid \beta_1, \ldots, \beta_p, \sigma^2, \lambda_2^2, \ldots, \lambda_p^2, \xi \sim \text{IG}\left(\frac{p}{2}, \frac{1}{2\sigma^2}\sum_{j=2}^{p}\frac{(\beta_j - \beta_{j-1})^2}{\lambda_j^2} + \frac{1}{\xi}\right),$$

$$\lambda_j^2 \mid \beta_j, \beta_{j-1}, \sigma^2, \tilde{\tau}^2, \nu_j \sim \text{IG}\left(1, \frac{(\beta_j - \beta_{j-1})^2}{2\sigma^2\tilde{\tau}^2} + \frac{1}{\nu_j}\right),$$

$$\nu_j \mid \lambda_j^2 \sim \text{IG}\left(1, \frac{1}{\lambda_j^2} + 1\right),$$

$$\xi \mid \tilde{\tau}^2 \sim \text{IG}\left(1, \frac{1}{\tilde{\tau}^2} + 1\right).$$

From the full conditional distributions, the Gibbs sampler can be constructed.

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